# EN3150 Assignment 03

Section 6: Final Comparison and Trade-off Discussion

Group Draco - *Dealing with AI from the core*

GUNARATHNA K.T.M.B. (230203G) JAYAWARDANA K.P.P. (230294H)
JITHMAL P.G.I. (230306B) NIMADITH W.N.A.C. (230442L)

## 1 Corrected Final Comparison

This comparison uses only leakage-free clean-split artifacts. Model B, MobileNetV2, and ShuffleNetV2 1.0x used the same 6,380/1,368/1,366 split. Checkpoint storage uses binary MiB; MACs count Conv2d and Linear operations for $1 \times 3 \times 64 \times 64$ input, with FLOPs equal to twice the MACs. Runtime RAM and inference latency were not measured.

Table 1: Canonical Model B versus pre-trained model comparison.

| Model | Test loss | Accuracy | Macro P | Macro R | Macro F1 | Parameters | Size | MACs | FLOPs |
|---|---|---|---|---|---|---|---|---|---|
| Model B | 0.6755 | 70.57% | 0.7416 | 0.6606 | 0.6800 | 37,957 | 0.176 MiB | 7.263 M | 14.526 M |
| ShuffleNetV2 1.0x | 0.3968 | 83.46% | 0.8705 | 0.8519 | 0.8606 | 1,258,729 | 4.966 MiB | 11.751 M | 23.502 M |
| MobileNetV2 | 0.3581 | 86.09% | 0.8824 | 0.8825 | 0.8822 | 2,230,277 | 8.740 MiB | 24.455 M | 48.910 M |

### 1.1 Accuracy

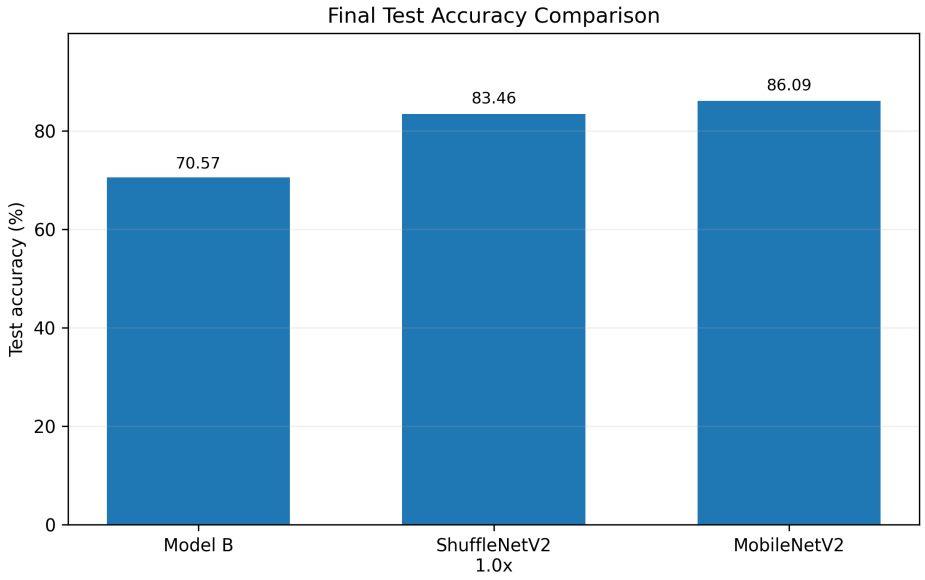
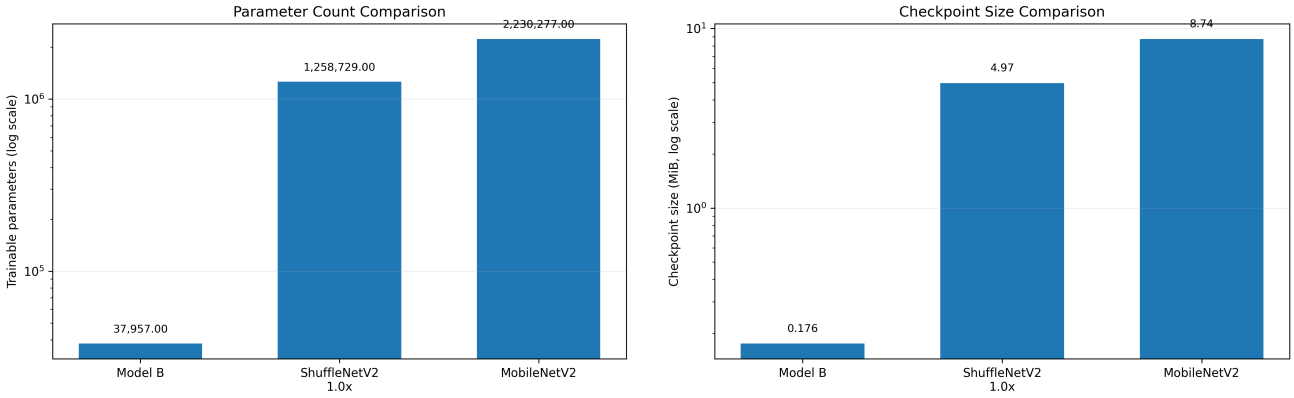



Figure 1: Leakage-free test accuracy comparison.

MobileNetV2 improves on Model B by 15.52 percentage points; ShuffleNetV2 improves by 12.88 points. ImageNet pre-training likely provides reusable visual features, but architecture and capacity also differ, so the full advantage cannot be attributed only to pre-training.

## 1.2 Parameter and Checkpoint Footprint




(a) Parameters.

(b) Checkpoint size.

Figure 2: Parameter and storage comparisons on logarithmic scales.

ShuffleNetV2 has 33.16× as many parameters and a 28.29× larger checkpoint than Model B. MobileNetV2 has 58.76× as many parameters and a 49.79× larger checkpoint. These file sizes do not determine peak runtime RAM, which also depends on activation buffers, operators, the framework, and allocation strategy.

## 1.3 Calculated Computational Cost

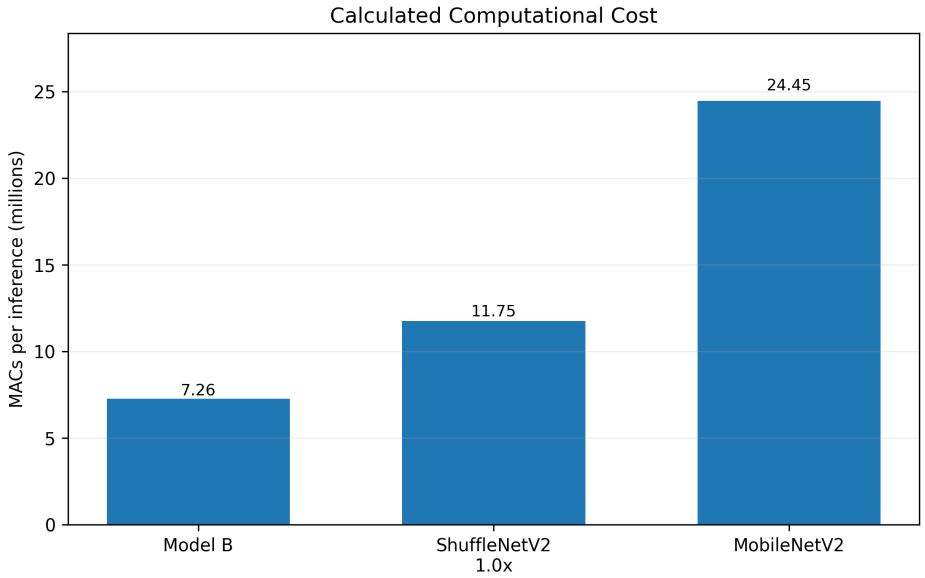
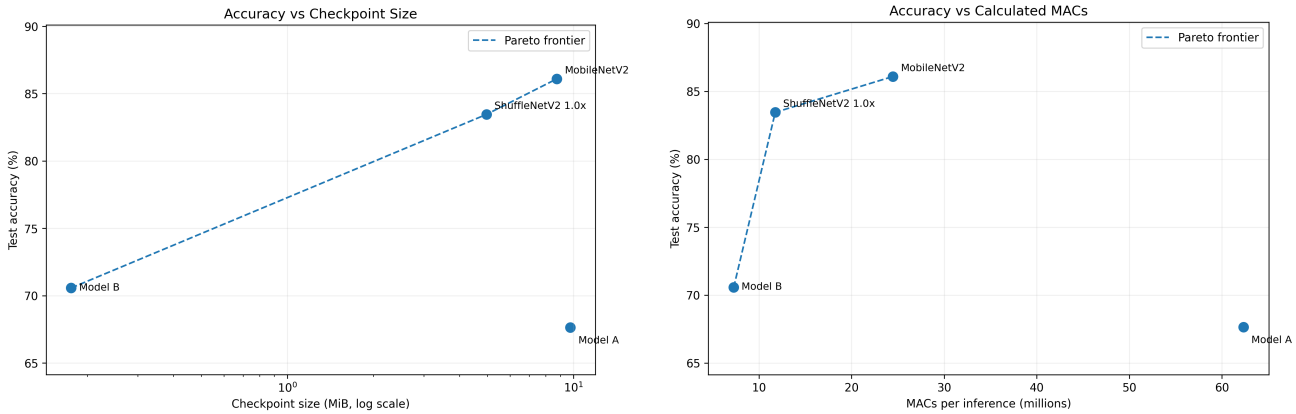



Figure 3: Calculated Conv2d/Linear MACs per input.

Model B needs 7.263 M MACs. ShuffleNetV2 uses 1.62× that amount, while MobileNetV2 uses 3.37×. These operation counts are not substitutes for target-device timing.

## 1.4 Accuracy-Efficiency Frontier




(a) Accuracy versus size.

(b) Accuracy versus MACs.

Figure 4: Leakage-free accuracy-efficiency trade-offs, including Model A as the conventional baseline.

Model A is dominated by Model B because Model B is both smaller and more accurate in this experiment. Model B, ShuffleNetV2, and MobileNetV2 offer distinct operating points: minimum resource cost, a middle accuracy-efficiency compromise, and maximum measured accuracy.

## 1.5 Engineering Recommendation

- **Strictest parameter/storage/compute budget: Model B.** Its depthwise-separable design is trained from scratch and contains only 37,957 parameters. Deployment still requires peak-RAM and latency measurement.
- **Balanced option: ShuffleNetV2 1.0x.** It reaches 83.46% accuracy while using fewer parameters, storage, and MACs than MobileNetV2.
- **Accuracy-first option: MobileNetV2.** It reaches 86.09% accuracy and macro F1 0.8822, with the largest arithmetic cost of the three.

# 2 Conclusion

The corrected experiments show a clear three-way trade-off. Model B minimizes resource demand, ShuffleNetV2 provides the middle point, and MobileNetV2 maximizes accuracy. The appropriate model depends on measured target-platform constraints rather than checkpoint size alone.